

Fourth-year Theoretical Physics student at Trinity College Dublin, focused on quantum thermodynamics, many-body dynamics, and computational physics. I build reproducible Python/PyTorch research tools that connect analytic models, numerical simulation, machine learning, and experimental data workflows.

Education

- 2022–2026

B.A. Theoretical Physics, Trinity College Dublin (expected)

Focus: quantum mechanics, statistical mechanics, quantum field theory, computational physics.
- 2020–2022

Leaving Certificate

The Dublin Academy of Education

Research And Physics Projects

- 2025–2026

Local Thermalty from Global Athermalty

Capstone thesis, supervisor: Felix Binder

 - Studied locally thermal bipartite quantum states with Gibbs marginals, isolating correlations as the remaining thermodynamic resource.
 - Characterized the LT set as an affine PSD-cone slice; derived a nine-parameter two-qubit normal form and Schur-complement positivity criterion.
 - Analysed diagonal slices, LT rays, commuting qutrit subclasses, and SDP tests for Gibbs-preserving convertibility using $D(\rho||\Gamma) = I(A : B)_\rho$ on LT.
- Jun–Sep 2025

Research Assistant, CRANN Institute

Trinity College Dublin

 - Designed Echo State Network reservoir models to learn quantum time-evolution data from simulated spin-chain observables.
 - Built a PyTorch ESN toolkit with configurable reservoirs, mixed precision, nine ridge-regression solvers, Optuna tuning, diagnostics, and Heisenberg-chain conservation checks.
- May–Sep 2024

Experimental / Computational Research Assistant

Trinity College Dublin

 - Created Python experiment-control and data-acquisition tooling for spectroscopy measurements on ferrofluids and liquid crystals.
 - Automated structured logging and repeatable measurement workflows; related work: arXiv:2504.19633.
- 2024–2025

DFT Hamilton / PySCF Work

Computational chemistry

 - Built a Dockerized PySCF workflow for H2 RHF/DFT, open-shell Li UHF, molecular examples, spin-correlation tests, endpoint checks, and custom functional comparisons.

Teaching And Technical Work

- Oct 2025–Present

Teaching Assistant

Trinity College Dublin

 - Lead weekly tutorials and problem-solving sessions; grade assignments and exams with clear feedback.
- 2024–2025

Software Engineer, EdTech

Grinds360

 - Built Python automation tools for content pipelines; contributed to product features and UX iteration with technical and non-technical stakeholders.

Technical Skills

Physics	Quantum thermodynamics, many-body dynamics, Gibbs-preserving maps, density matrices, SDP methods, spectroscopy
Computing	Python, NumPy, SciPy, PyTorch, QuTiP, Matplotlib, Mathematica, C/C++, Docker, SQLite, Git/GitHub
Research practice	LaTeX, numerical simulation, data analysis, hyperparameter tuning, reproducible pipelines, technical writing
Languages	English (fluent), Russian (intermediate), Spanish (basic)